

Nikhilesh Vijay Chirutala

Full-Stack Software Engineer | Backend Software Engineer

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SUMMARY

Backend-focused Full-Stack Software Engineer with 3 years of experience building and maintaining production applications across cloud, CRM, AI, and contact-center platforms. Experienced in designing REST APIs, backend services, business logic, data models, and cloud-based workflows using Python, FastAPI, Django, MongoDB, PostgreSQL, and AWS. Strong full-stack experience with React.js and Next.js, delivering end-to-end product features across frontend, backend, APIs, and data layers. Experienced in production troubleshooting, reliability improvements, authentication and authorization, testing, and integration of external services.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3

Backend & Frameworks: Python, FastAPI, Django, Flask, REST APIs, API Development, Backend Services, Backend Automation

Frontend: React.js, Next.js, Tailwind CSS, Bootstrap, Jinja2

Databases: MongoDB, PostgreSQL, MySQL, Neon DB, SQL, Data Modeling

Cloud & DevOps: AWS, Amazon S3, AWS Lambda, Vercel, GitHub Actions, CI/CD, Linux

APIs & Integrations: REST APIs, API Integration, Twilio, GitHub API, WebSockets, Postman

Security & Authentication: Authentication, Authorization, RBAC, ACL, JWT

Testing & Engineering: Unit Testing, Jest, Postman, Git, GitHub, Debugging, Code Reviews, SonarQube

AI & LLM: RAG, LLM Integration, Vector Search, Prompt Engineering

WORK EXPERIENCE

Software Engineer • CA One Tech Cloud Inc.

Apr 2026 – Present | Bengaluru, India

Client: Convoso — Cloud Contact Center Platform

Python, FastAPI, AWS, Amazon S3, MySQL, Datadog, Grafana

- Develop and maintain production-grade Python backend services for a high-volume cloud contact center platform, supporting data processing, call-recording workflows, storage operations, and production use cases.
- Migrated a legacy Perl-based workflow to a long-running Python service deployed across 40 production clusters, supporting staged rollout, feature flags, validation, rollback, and production monitoring.
- Built Python service integrations with a Call Intelligence microservice, implementing APIs, eligibility checks, account and campaign controls, processing limits, retries, response validation, and result persistence.
- Developed validation and recovery workflows for production data, handling incomplete processing, failed operations, inconsistent storage states, and recovery scenarios across multiple environments.
- Built AWS S3 audit and recovery workflows for recording storage, implementing deterministic object handling and idempotent re-upload operations for incomplete or failed storage workflows.
- Supported production deployments across multiple clusters through configuration, staged validation, monitoring, troubleshooting, and rollback activities.
- Implemented production observability using Datadog, Grafana dashboards, and Sentry to monitor service health, diagnose failures, and support production incident resolution.
- Investigated and resolved production issues across Python services, APIs, databases, and cloud storage while improving reliability of recording-processing workflows.

Software Engineer • Mantra Technologies

Oct 2023 – Mar 2026 | Hyderabad, India

- Delivered production features across multiple applications by translating product requirements into backend services, REST APIs, business logic, data workflows, and end-to-end product functionality using Python, FastAPI, and modern JavaScript frameworks.
- Developed and integrated application functionality across frontend, backend, API, and data layers, working with React.js, Next.js, MongoDB, and relational databases to implement data models, CRUD workflows, relationships, validation, and data-driven features.
- Built secure application workflows by implementing authentication, authorization, RBAC, ACL, and JWT-based access control, while integrating external services and resolving API and integration-related issues.
- Supported production releases and maintained application reliability through unit testing, debugging, root-cause analysis, deployment validation, code reviews, defect resolution, and continuous improvements to backend and full-stack workflows.

Key Projects:

AI Bull — AI-Powered Trading & Investment Platform

React.js, Next.js, FastAPI, Python, MongoDB, RAG, LLM

- Developed an AI-powered trading and investment platform with portfolio aggregation, enabling users to consolidate holdings from multiple Demat accounts into a unified application.
- Built backend APIs and business logic for portfolio aggregation, investment tracking, stock and mutual fund analysis, screening, reporting, and backtesting workflows.

- Designed data-driven workflows for tracking positions, stocks, mutual funds, portfolio information, and investment-level reports across aggregated accounts.
- Developed configurable stock and mutual fund screeners with filtering capabilities and detailed asset pages for analyzing individual investments.
- Designed and implemented a backtesting engine for evaluating trading strategies against historical market data and analyzing strategy performance.
- Implemented a Retrieval-Augmented Generation (RAG) chatbot that combines portfolio and financial information to provide contextual answers to portfolio-related user queries.

CRM.io — AI-Driven CRM System

FastAPI, Python, Jinja2, MongoDB, ACL, Twilio, WebSockets, AWS S3, AWS Lambda, FCM

- Developed backend modules and REST APIs using Python and FastAPI for Contacts, Tasks, Deals, Documents, communication, and account-management workflows.
- Built contact-management workflows with inline editing, global search, related Tasks, Deals and Documents, and bulk contact import through Google Sheets integration.
- Developed a real-time CRM dialer using Twilio and WebSockets, integrating telephony workflows with contact records and application-level communication features.
- Implemented CRM-wide soft-delete and Trash workflows with a 30-day retention period, using an AWS Lambda service for permanent record cleanup.
- Built document-management workflows using Amazon S3, including browser-to-S3 uploads and storage of documents associated with CRM records.
- Developed task cadence workflows allowing users to create and execute sequences of scheduled follow-ups, with Firebase Cloud Messaging for notifications.
- Built account-level reporting dashboards and an MCP integration exposing CRM capabilities as tools for AI-assisted retrieval and supported operations.

PERSONAL PROJECTS

JS Studio — Online JavaScript Playground

Next.js, GitHub API, Vercel

- Developed an online JavaScript playground for writing and executing JavaScript through a web interface.
- Integrated GitHub APIs and deployed the application using Next.js and Vercel.

AnyScrum — Project Management Platform

Next.js, Neon DB, Clerk, Vercel

- Developed a project management platform supporting projects, tasks, and team workflows.
- Implemented authentication and user management with Clerk and persistent application data with Neon DB.

Enterprise REST API Platform — Django and Flask Backend Application

Python, Django, Django REST Framework, Flask, PostgreSQL, REST APIs

- Built modular REST APIs using Django, Django REST Framework, and Flask for authentication, business logic, and data-driven workflows.
- Designed PostgreSQL data models with CRUD operations, relationships, validation, and API error handling.
- Implemented JWT authentication and role-based access control, with API testing and validation using Postman.

EDUCATION

B.Tech — Electrical & Electronics Engineering

2019 – 2023 | GPA: 7.73 / 10

Srinivasa Ramanujan Institute of Technology, Anantapur